Agent task: Daytona Vera versus Daytona zen5

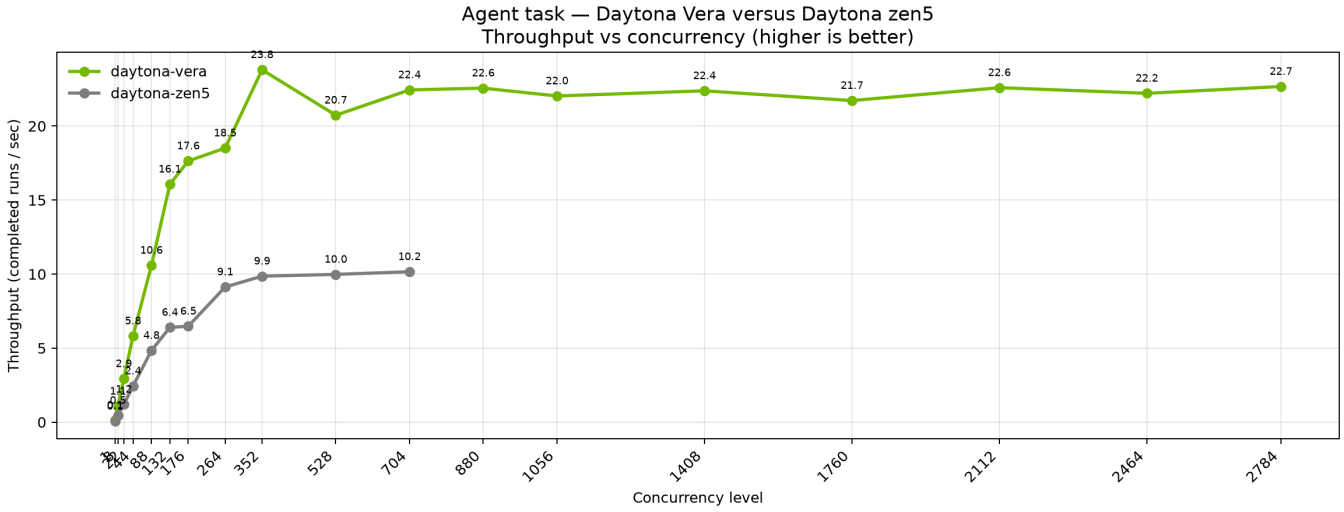
August 2026. Coding-agent sandbox results on NVIDIA Vera and AMD EPYC Zen 5 (Phoenix).

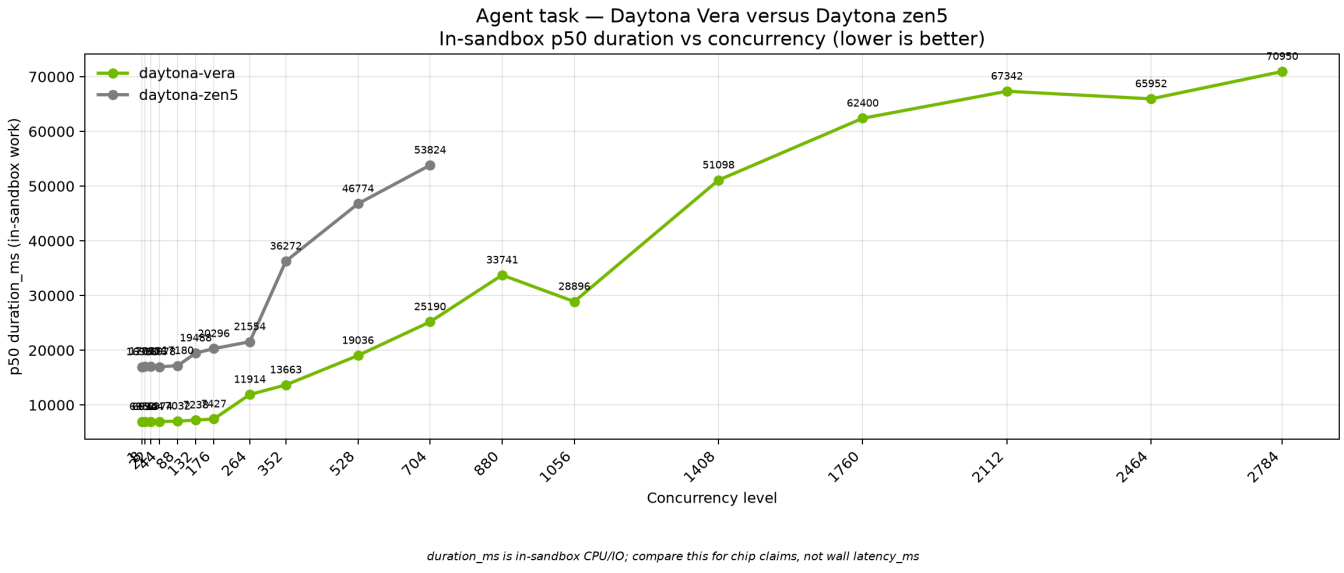
This brief covers one workload: the **agent task** Daytona uses to stand in for coding-agent sessions inside isolated sandboxes. Both chips ran the same image, the same seed, and the same concurrency ladder.

What to take away

On a single sandbox, Vera finished the agent loop in about **7.0 seconds** median. Zen5 took about **17.0 seconds**. That is roughly **2.4× faster** in-sandbox time on Vera.

As concurrency climbs, Vera keeps packing more completed jobs per second. Throughput reaches about **23.8 jobs/s** at 352 concurrent sandboxes and stays near **23 jobs/s** through **2,784** concurrent sandboxes on Vera. Zen5 plateaus near **10 jobs/s** through 704. In-sandbox median time stays shorter on Vera at every level we measured.





Why this matters for Daytona (coding-agent RL)

Daytona's product story is not only "faster sandboxes." It is that **coding-agent reinforcement learning turns sandboxes into part of the training loop**. Agents inspect repos, edit code, run commands, and re-test inside isolated environments; those rollouts feed policy / reward training on expensive accelerators. When the CPU-side environment is slow to start or slow to finish, **GPU trainers wait** — the classic RL "starve the cluster" failure mode.

We measured that **rollout infrastructure tax** in a companion study comparing four execution substrates (single containers, hosted sandboxes, Kubernetes-orchestrated containers, and cloud VMs) under the same coding-agent workloads (arXiv:2607.01415, *The Rollout Infrastructure Tax in Coding-Agent Reinforcement Learning*, SoCC 2026 submission). Headline results from that paper:

- Cold-start latency varied by up to **110×** across substrates.
- For **one million** 150-step trajectories, substrate choice produced a **1.8×** spread in projected rollout worker-hours — about **5,316 extra worker-hours** between the slowest and fastest substrate (42.5 s vs 23.4 s per trajectory).
- Even **1 second** of extra overhead per rollout is **278 worker-hours** at one million trajectories; **100 ms** per action across 150 steps is **4,167 worker-hours**.

Those numbers compare sandbox substrates. This brief asks the next question: once you are on Daytona, how much faster is Vera than Zen5 for each coding-agent episode?

Example: Vera's chip gain at RL scale

On this agent task, Vera completes an idle episode in about **7.0 s** vs about **17.0 s** on Zen5 (**~2.4×** faster), and at 352 concurrent sandboxes delivers about **23.8 jobs/s** vs about **9.9 jobs/s** (**~2.4×** higher packing rate).

Using the same 1M-trajectory projection style as the paper — worker-hours = (number of episodes × seconds per episode) / 3600, excluding model inference:

Chip	Median episode time (this brief, c=1)	Projected worker-hours for 1M episodes
Zen5	~17.0 s	**~4,710 h**
Vera	~7.0 s	**~1,940 h**
Saved on Vera	~10 s / episode	**~2,770 worker-hours**

That is the same compounding math as the white paper's substrate gap: a few seconds per coding-agent episode becomes thousands of worker-hours at post-training scale. Faster episodes also mean **rollout batches arrive sooner to the GPU side**, so accelerators spend less time idle waiting for environments — the infrastructure tax the paper ties to underutilized training hardware.

Illustrative GPU cost (example only). If a training run keeps even a modest accelerator pool waiting on sandbox rollouts — say the equivalent of **8× H100-class GPUs at ~$3/GPU-hour** while rollouts are the bottleneck — reclaiming utilization proportional to a **~2.4×** faster sandbox completion rate is on the order of **tens of thousands of dollars** over a multi-day RL run, before counting the CPU worker-hour savings above. Exact dollars depend on cluster size, sync vs async RL, and how often GPUs actually stall on rollouts; the measured chip result is the **~2.4×** in-sandbox completion advantage on Vera.

The agent task

Daytona customers run coding agents inside sandboxes: open a project, search the tree, read and parse code, edit files, then verify with tests. This agent task mirrors that loop without calling an LLM and without leaving the sandbox.

Each episode:

1. **Seeds** a multi-file Python package with deliberate bugs and a large generated module tree.
2. **Searches** the workspace for symbols and call sites (the same kind of repo walk a coding agent does).
3. **Parses** modules with the AST so the work is real Python analysis, not a toy loop.
4. **Applies** deterministic patches that repair the seeded bugs.
5. **Runs a heavy pytest suite** (parametrized CPU-burning cases) so most of the wall time is in-sandbox compute and filesystem I/O—the same cost profile we see when agents edit and re-test code in Daytona.

Successful episodes return a matching checksum on both chips, so Vera and Zen5 completed the same work.

How to read the charts

Throughput (jobs per second) is completed sandbox episodes divided by the wave's exec wall clock. Higher is better. It answers: how many isolated agent sessions can the platform finish per second as we pack the machine?

p50 duration_ms is the median time spent *inside* the sandbox doing the agent loop. It excludes sandbox create, delete, and client network. Lower is better. It is the chip-facing metric for "how long does one agent episode take?"

Configuration

Setting	Value
Workload	Agent coding loop (`repo-agent-v3`)
Snapshot	`dtgraviet/vera-agent-benchmark:v3` (multi-arch amd64 + arm64)
Scale factor `--n`	**50** (calibrated for multi-second idle work)
Seed	**42**
Episodes per sandbox (`-E`)	**8**
Concurrency ladder	1–704 (1 GiB); Vera max-pack extension 880–2,784 (512 MiB)
CPU guarantee	**0.125** vCPU per sandbox
CPU burst ceiling	**1.0** vCPU (`--rlp-cpu-max 1`)
Memory / disk	**1 GiB** memory through 704; Vera max-pack uses **512 MiB** (`--rlp-memory 0.5`) and **2 GiB** disk
Vera cell	On-node client next to the Vera RLP runner (`aarch64`)
Zen5 cell	Phoenix (`us-phoenix-1`, AMD EPYC / `x86_64`)

Matched recipe on both sides (base ladder, 1 GiB):

```
UV_NO_SYNC=1 uv run main.py --benchmark agent --runner rlp \
  --snapshot dtgraviet/vera-agent-benchmark:v3 \
  --levels 1 8 22 44 88 132 176 264 352 528 704 \
```

```
  --n 50 --seed 42 -E 8 --hold-then-exec \
  --rlp-cpu 0.125 --rlp-cpu-max 1
```

Vera max-pack extension (512 MiB, same CPU burst):

```
UV_NO_SYNC=1 uv run main.py --benchmark agent --runner rlp \
  --snapshot dtgraviet/vera-agent-benchmark:v3 \
  --levels 704 880 1056 1408 1760 2112 2464 2784 \
  --n 50 --seed 42 -E 8 --hold-then-exec \
  --rlp-cpu 0.125 --rlp-cpu-max 1 --rlp-memory 0.5
```

Vera used `--target vera`. Zen5 used `--target us-phoenix-1`.

The **0.125 / max 1** CPU settings let both cells pack far past a hard 1-vCPU-per-sandbox reservation, so the high concurrency rungs (352–704) are a fair density compare rather than an artificial capacity cliff on either side.

Headline numbers

Concurrency	Vera p50 duration	Zen5 p50 duration	Vera throughput	Zen5 throughput
1	6,952 ms	16,960 ms	0.13 /s	0.06 /s
88	7,032 ms	17,180 ms	10.59 /s	4.84 /s
176	7,427 ms	20,296 ms	17.64 /s	6.48 /s
352	13,663 ms	36,272 ms	23.80 /s	9.86 /s
704	25,190 ms	53,824 ms	22.44 /s	10.16 /s
1,760	62,400 ms	—	21.72 /s	—
2,784	70,950 ms	—	22.67 /s	—

Both Vera and Zen5 finished with **zero failed jobs** through 704 on the 1 GiB ladder. Vera max-pack (512 MiB) held **0 fails** through **1,760**; at **2,112+** create failures appear as the live-VM cap is reached (~2,045 sandboxes), but throughput stays near **23 jobs/s**.

Method notes

- Duration figures are median in-sandbox `duration_ms`. Throughput uses completed runs over measured exec wall.
- Both series use the same snapshot tag, seed, `--n`, episode count, and concurrency ladder.
- Each cell is one machine. Absolute packing at the top of the ladder also reflects how many sandboxes that cell can keep live; prefer duration for chip claims and throughput for packing.
- RL scale projections reuse the worker-hour formula from Graviet et al., arXiv:2607.01415; GPU dollar figures are illustrative, not a measured billing study.
- Charts in this brief: `eda_output/nvidia-brief-agent/` (regenerate with `uv run python scripts/nvidia_brief_agent_charts.py`).
- Source JSONL:
- Vera (1 GiB): `data/agent/rlp-vera-c0p125-max1/concurrency_20260826_005637_n50.jsonl`
- Vera (512 MiB max-pack): `data/agent/rlp-vera-c0p125-max1-m512/concurrency_20260826_230252_n50.jsonl`
- Zen5 (1 GiB): `data/agent/rlp-phoenix-c0p125-max1/concurrency_20260826_012143_n50.jsonl`